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Artificial Intelligence and Audit Quality in the Banking Sector: Empirical Evidence From a Developing Economy

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Abstract: Artificial intelligence is increasingly being adopted across financial institutions to enhance operational efficiency, strengthen risk management, and improve the quality of assurance services. However, empirical evidence regarding its influence on audit quality in developing economies remains limited. This study investigated the impact of artificial intelligence adoption on audit quality within the commercial banking sector of Zimbabwe. A quantitative research design was employed, and primary data were collected through structured questionnaires administered to auditors and managerial personnel working in commercial banks. The findings indicate that the integration of artificial intelligence technologies into audit processes is associated with significant improvements in audit quality. Specifically, the quality and reliability of audit evidence were reported to be enhanced, the likelihood of material misstatements was perceived to be reduced, and greater efficiency in audit execution and decision-making was achieved. Despite these benefits, several barriers to implementation were identified, including inadequate technological infrastructure, limited financial capacity for artificial intelligence investment, and shortages of personnel with specialized artificial intelligence-related competencies. Nevertheless, strong support for the adoption of artificial intelligence-based auditing practices was observed among respondents. The results suggest that artificial intelligence has considerable potential to enhance audit quality by improving the accuracy, consistency, and reliability of audit procedures and evidence evaluation. It is therefore recommended that commercial banks increase investment in artificial intelligence-enabled audit technologies. Furthermore, supportive regulatory frameworks, professional standards, and implementation guidelines should be established by policymakers and financial regulators to facilitate responsible artificial intelligence adoption, mitigate emerging risks, and promote consistency in audit practices across the banking sector. These findings contribute to the growing body of literature on artificial intelligence-driven auditing and provide practical insights for financial institutions operating in developing economies.

Keywords: Artificial intelligence; Audit quality; Commercial banks in Zimbabwe; Technological challenges; Regulatory compliance

1. Introduction

In recent years, the adoption of artificial intelligence has gained momentum across various sectors, including banking and auditing. As financial transactions grow in complexity and volume, there is increasing pressure on auditors to enhance the efficiency, accuracy, and reliability of their work. Traditional auditing methods, which often rely heavily on manual processes and professional judgment, are no longer sufficient to meet the demands of modern financial systems. Artificial intelligence technologies such as machine learning, natural language processing, and robotic process automation are being integrated into audit processes to address these challenges. These tools enable auditors to analyze large datasets, detect anomalies, improve risk assessments, and ultimately enhance audit quality. In developing countries, where resource constraints and limited technological infrastructure can pose barriers, the potential benefits of artificial intelligence in auditing are particularly significant.

This study explores the impact of artificial intelligence on audit quality within the context of commercial banks

in a developing country, specifically Zimbabwe. It aims to determine whether artificial intelligence enhances audit evidence, reduces material misstatements, and improves overall audit outcomes. This study employs a quantitative research approach, using questionnaires distributed to auditors and bank managers. The findings reveal that while artificial intelligence offers clear benefits in improving audit quality, its adoption is hindered by challenges such as limited expertise and lack of technological readiness. The study underscores the importance of investing in artificial intelligence capabilities and recommends regulatory support to facilitate its effective integration into the auditing practices of the banking sector.

2. Literature Review

2.1 Historical Evolution of Auditing Practices

The audit profession was first introduced for checking activities in ancient civilizations and has gone through a lot of transformation over the years. However, auditing practice became known during the Industrial Revolution between 1800 and 1900 when auditing was about ensuring the correctness of accounts and detecting fraud and errors. In the present day, the services of auditors are expected to provide value-added services beyond expressing an audit opinion on the financial statements (Heang et al., 2019). According to the Association of International Chartered Certified Public Accountants, during these periods of auditing transformation, audit practices were not conducted independently; instead, auditors relied on information from management to formulate their audit opinions and this contributed to audit scandals (Heang et al., 2019). The scandals such as Enron, WorldCom, Xenon, and many other accounting and auditing practices made the news, contributing to the enactment of laws and regulations (Heang et al., 2019). The Sarbanes-Oxley Act was enacted in response to major accounting scandals and auditing failures, with the aim of deterring future misconduct and restoring investor confidence.

2.2 Role of Artificial Intelligence in Enhancing Audit Quality

Artificial intelligence has been transforming various industries, and the field of audit quality is no exception. The integration of artificial intelligence in audit quality has been driven by the need for more efficient, accurate, and reliable audits and the development of artificial intelligence in audit quality can be traced back to the 1950s, when computer scientists and mathematicians began exploring the possibilities of automating complex tasks (Ahmad, 2019). Over the years, artificial intelligence technology has evolved, and its applications have expanded to various industries, including finance and auditing. In the context of audit quality, artificial intelligence has been utilized to address the challenges associated with manual auditing processes (Heang et al., 2019). These challenges include the need for more efficient data analysis, the reduction of human errors, and the minimization of subjectivity in audit decision-making (Ahmad, 2019). Over the years, artificial intelligence has been successfully implemented in audit quality improvement in banks across various countries (Dagilienė & Klovienė, 2019). The implementation of artificial intelligence has enhanced the efficiency and accuracy of audits, enabling financial institutions to improve risk management and maintain regulatory compliance.

2.3 Global Trends and Regional Perspectives on Artificial Intelligence Adoption in Auditing

Across the European Union, regulatory bodies have also been encouraging the adoption of artificial intelligence in audit quality (Salijeni et al., 2019). The European Securities and Markets Authority has published guidelines on the use of artificial intelligence and other technologies in audits, promoting the benefits of increased accuracy and efficiency (Raghunandan, 2021). Numerous audit firms throughout the European Union member states have implemented artificial intelligence-powered tools to enhance audit quality and ensure regulatory compliance within European banks. The big four accounting firms, such as Deloitte, Ernst & Young, KPMG, and PwC, have incorporated artificial intelligence into their audit processes to improve the efficiency and accuracy of banks (Seethamraju & Hecimovic, 2023). Artificial intelligence has been making significant strides in various industries, and the banking sector in Africa is no exception (Rahman et al., 2021). Over the past two decades, accounting firms in Zimbabwe have incorporated artificial intelligence into their audit processes. However, they continue to face challenges related to the seamless adoption and integration of artificial intelligence technologies.

Over the course of the past years, accounting firms have come up against a number of obstacles in relation to the auditing of financial statements, such as the need for more efficient data analysis, the reduction of human errors, and the minimization of subjectivity in audit decision-making which led to organizations receiving a qualified opinion (Celestin, 2020). According to Jenkins & Stanley (2019), the methods that accounting firms use to adopt and implement information technology in auditing are ineffective due to the fact that many firms lack any clear knowledge regarding the strategies that can be used to enhance audit quality. Research on auditing practices in Zimbabwe has yet to clearly explain the reasons behind accounting firms' difficulties in successfully adopting artificial intelligence.

Specifically, prior studies have not adequately examined three critical dimensions within the Zimbabwean banking context: (i) the extent to which artificial intelligence tools such as machine learning and robotic process automation measurably improve audit evidence quality and reduce material misstatements in commercial banks; (ii) the specific institutional, regulatory, and skills-related barriers that prevent Zimbabwean banks from successfully integrating artificial intelligence into their audit functions; and (iii) the degree to which artificial intelligence adoption translates into verifiable improvements in audit reliability and compliance outcomes under the unique socioeconomic and infrastructural constraints of a developing country. While global and African regional studies provide broad insights, none has quantitatively investigated all three dimensions simultaneously within the Zimbabwean commercial banking sector. This study therefore addresses these underexplored aspects by providing empirical evidence from Zimbabwe, thereby contributing to the broader discourse on artificial intelligence-driven audit quality in developing countries.

A comparative review of the literature reveals important methodological and contextual divergences that the preceding sections do not adequately foreground. Studies conducted in developed economies rely predominantly on qualitative case studies and regulatory analysis, whereas African-focused work employs descriptive and exploratory designs with limited empirical measurement of audit outcomes (Omoteso, 2012). Notably, while prior studies affirm artificial intelligence's capacity to enhance audit evidence quality, none isolates the effect of specific artificial intelligence tools (e.g., machine learning versus robotic process automation) on distinct audit quality dimensions—a distinction that matters considerably in resource-constrained environments like Zimbabwe. Furthermore, studies that identify adoption barriers do not compare institutional responses across bank types or regulatory contexts, leaving unresolved whether the challenges are sector-wide or firm-specific (Seethamraju & Hecimovic, 2023)—a gap this study directly addresses through quantitative data from multiple Zimbabwean commercial banks.

2.4 Adoption of Artificial Intelligence in African Banking Audits: The Case of Zimbabwe

Artificial intelligence has been implemented in various aspects of audit quality in African banks. For instance, machine learning algorithms have been used to analyze large volumes of transaction data to detect patterns and anomalies that may indicate fraud or mismanagement (Omoteso, 2012). Natural language processing has been employed to automate the process of reviewing and summarizing large volumes of financial documents, enabling auditors to focus on higher-value tasks (Omoteso, 2012). Furthermore, robotic process automation has been implemented to automate routine and repetitive tasks like data entry and report preparation, allowing auditors to focus more on complex and value-added analyses. The adoption of artificial intelligence in audit quality by banks in Africa has the potential to significantly improve the efficiency and effectiveness of audit processes (Chukwuani & Egiyi, 2020). By automating routine tasks and enabling auditors to focus on higher-value activities, artificial intelligence can help banks identify and address risks more effectively, thereby reducing the likelihood of financial scandals and regulatory penalties (Ibrahim & Jahswill, 2025). Moreover, artificial intelligence assists banks in staying compliant with changing regulatory standards while enhancing the accuracy and transparency of their financial reporting.

Artificial intelligence has made significant strides in various industries, including banking and finance. In Zimbabwe, banks have been exploring the use of artificial intelligence to improve audit quality and streamline their operations (Jachi, 2019). The use of artificial intelligence by banks in Zimbabwe can be traced back to the early 2000s when the country's banking sector began to adopt advanced technology to improve efficiency and reduce human error (Jachi, 2019). The adoption of artificial intelligence in banking has been motivated by factors such as the necessity to comply with international banking regulations, the growing complexity of financial transactions, and increasing demands for transparency and audit accuracy. Against this backdrop, this study aims to assess the impact of artificial intelligence on audit quality, focusing on commercial banks in Zimbabwe.

2.5 Profile of Selected Commercial Banks in Zimbabwe

Currently, Zimbabwe's banking sector comprises 19 licensed institutions, including 14 commercial banks, 4 building societies, and 1 merchant bank. This study randomly selected four commercial banks, representing approximately 30% of the total commercial banking population. Based on the guideline by Wright (2019), the study selected four commercial banks—CBZ, FBC, Nedbank, and First Capital Bank—representing approximately 30% of Zimbabwe's commercial banking sector. CBZ was established in January 1998, while FBC began operations in August 1997. First Capital Bank was founded in 1912 as the Bank of Africa, later becoming Barclays, and ultimately rebranding as First Capital Bank. Nedbank commenced operations in Zimbabwe in 2003 as MBCA before rebranding to Nedbank in 2018. According to Shambira (2020), Zimbabwean banks are gradually adopting artificial intelligence, driven by technological advancements and the need to improve operational efficiency.

2.6 Research Question

The research question of this study is: "What is the impact of artificial intelligence on audit quality in developing countries?"

3. Methodology

This study employed a quantitative research approach to investigate the impact of artificial intelligence on audit quality in Zimbabwe's commercial banking sector. The quantitative design was chosen to allow for the collection of numerical data that could be objectively measured and analyzed. Data were collected using structured questionnaires, which were distributed to two target groups: bank managers and auditors. The questionnaire focused on key areas such as the effectiveness of artificial intelligence in improving audit evidence, reducing material misstatements, enhancing compliance, and overcoming operational challenges in the auditing process.

The questionnaire items were specifically developed for this study to reflect the unique technological and institutional context of Zimbabwe's commercial banking sector, drawing on the conceptual themes identified in the literature review rather than being directly adapted from a single pre-existing validated instrument. Each construct—comprising four items per dimension measured on a five-point Likert scale—was designed to capture respondents' perceptions of artificial intelligence's effect on audit evidence quality, material misstatement risk, audit reliability, and adoption challenges, with content validity assured through alignment with the study's objectives and the internal consistency of responses reflected in the descriptive statistics reported in Tables 1, 2, and 3.

As shown in Table 1, a total of 37 bank managers participated in the study. The sample size was determined based on Morgan and Krejcie table in Chuan & Penyelidikan (2006), which provides a scientifically grounded method for selecting appropriate sample sizes from known populations. Additionally, five auditors were purposively selected to provide professional insights on the implementation and effectiveness of artificial intelligence in audit functions. Their input added depth and technical perspective to the findings. The use of questionnaires ensured consistency in data collection, while the analysis was conducted using descriptive statistics to identify trends, summarize responses, and draw meaningful conclusions about the role of artificial intelligence in audit quality.

Table 1. Population and sample size

Participants	Population	Sample
Branch managers	42	37
Auditors	5	5

It is acknowledged, however, that the disparity in sample size between the 37 bank managers and the 5 auditors introduces a potential imbalance in the data, as the findings are weighted more heavily towards managerial perspectives than towards the technical viewpoints of audit practitioners. This imbalance may limit the extent to which the results fully capture the operational realities of artificial intelligence adoption in audit practice, and readers should interpret the auditor-specific findings with corresponding caution. Future studies are encouraged to incorporate a more proportionate representation of auditors to strengthen the comparability and generalizability of conclusions across both groups.

4. Data Analysis and Presentation

4.1 Descriptive Statistics

Table 2 presents the descriptive statistics for the key variables used in the study. Each construct consisted of four items, which were combined into new variables for analysis. The variable "impact of artificial intelligence on audit evidence quality" (AIAV_MEAN) had 35 responses, with a mean score of 4.1889, indicating general agreement on the Likert scale, and a standard deviation of 0.53. The "impact of artificial intelligence on risk of material misstatement" (AIMM_MEAN) also had 35 responses, showing a mean of 4.0333 and a standard deviation of 0.73. For the "impact of artificial intelligence on reliability of audit evidence" (AIRAE_MEAN), the mean was 3.7889, with a standard deviation of 0.77, based on 35 observations. Lastly, the "challenges associated with artificial intelligence in audit quality" (CAIOAQ_MEAN) had a mean of 3.98 and a standard deviation of 0.48. These results suggest that most participants agreed with the statements under each construct. Overall, the descriptive statistics indicate that the data for each variable is normally distributed, and the constructs used in the study demonstrate strong reliability and consistency.

Table 2. Descriptive statistics

Variables	*N*	Minimum	Maximum	Mean	Standard Deviation
AIAV_MEAN	35	2.25	5.00	4.1889	0.53075
AIMM_MEAN	35	1.75	5.00	4.0333	0.72809
AIRAE_MEAN	35	1.25	4.75	3.7889	0.77415
CAIOAQ_MEAN	35	2.50	4.75	3.9833	0.48383
Valid N (listwise)	35	-	-	-	-

Note: *N* = sample size; AIAV_MEAN = impact of artificial intelligence on audit evidence quality; AIMM_MEAN = impact of artificial intelligence on risk of material misstatement; AIRAE_MEAN = impact of artificial intelligence on reliability of audit evidence; CAIOAQ_MEAN = challenges associated with artificial intelligence in audit quality. "-" indicates not applicable.

4.2 Correlation Matrix

Table 3 presents the correlation matrix for the key study variables: AIAV_MEAN, AIMM_MEAN, AIRAE_MEAN, and CAIOAQ_MEAN. The sample size for this analysis was $n = 35$. At the 5% significance level, the results indicate several notable relationships. There is a moderate positive correlation between AIAV_MEAN and AIMM_MEAN, with a coefficient of 0.417, suggesting a statistically significant association between these two variables. The correlation between AIRAE_MEAN and AIAV_MEAN was 0.168, indicating a weak positive relationship. Meanwhile, CAIOAQ_MEAN and AIAV_MEAN showed a stronger positive correlation, with a coefficient of 0.538, also statistically significant.

Table 3. Correlation matrix

Variables	Statistics	AIAV_MEAN	AIMM_MEAN	AIRAE_MEAN	CAIOAQ_MEAN
AIAV_MEAN	Pearson Correlation	1	0.417**	0.168	0.538**
	Sig. (2-tailed)	-	0.004	0.269	0.000
	N	35	35	35	35
AIMM_MEAN	Pearson Correlation	0.417**	1	0.318*	0.433**
	Sig. (2-tailed)	0.004	-	0.033	0.003
	N	35	35	35	35
AIRAE_MEAN	Pearson Correlation	0.168	0.318*	1	0.628**
	Sig. (2-tailed)	0.269	0.033	-	0.000
	N	35	35	35	35
CAIOAQ_MEAN	Pearson Correlation	0.538**	0.433**	0.628**	1
	Sig. (2-tailed)	0.000	0.003	0.000	-
	N	35	35	35	35

Note: *N* = sample size; Sig. = significance; AIAV_MEAN = impact of artificial intelligence on audit evidence quality; AIMM_MEAN = impact of artificial intelligence on risk of material misstatement; AIRAE_MEAN = impact of artificial intelligence on reliability of audit evidence; CAIOAQ_MEAN = challenges associated with artificial intelligence in audit quality. ** indicates that correlation is significant at the 0.01 level (2-tailed), and * indicates that correlation is significant at the 0.05 level (2-tailed). "-" indicates not applicable.

Additionally, the correlation between AIMM_MEAN and AIRAE_MEAN was 0.318, reflecting a weak positive relationship, while AIMM_MEAN and CAIOAQ_MEAN had a coefficient of 0.433, also suggesting a moderate positive relationship. The strongest observed relationship was between CAIOAQ_MEAN and AIRAE_MEAN, with a correlation coefficient of 0.628, indicating a significant positive relationship. Overall, these results highlight that while some variables are only weakly correlated, others demonstrate more meaningful associations in the context of artificial intelligence adoption and audit quality.

4.3 Effect of Artificial Intelligence on the Credibility and Accuracy of Audit Findings

Figure 1 illustrates the responses gathered from bank managers regarding whether artificial intelligence enhances the accuracy of audit evidence. The results show that 43% of respondents strongly agreed, while 31% agreed, indicating that a combined 74% of the bank managers supported the view that artificial intelligence contributes to more accurate audit evidence. Meanwhile, 11% of the participants remained neutral, and 15% disagreed with the statement. These findings suggest a generally positive perception of artificial intelligence's effectiveness in improving the reliability of audit evidence. This aligns with the conclusions of Tapscott & Tapscott (2018), who found that artificial intelligence has significantly transformed the way audit evidence is collected, analyzed, and interpreted—ultimately enhancing the overall quality of audits. Their research emphasized that artificial intelligence's capacity to process vast amounts of data rapidly and accurately surpasses human ability, thus reinforcing the value of artificial intelligence in producing high-quality audit outcomes.

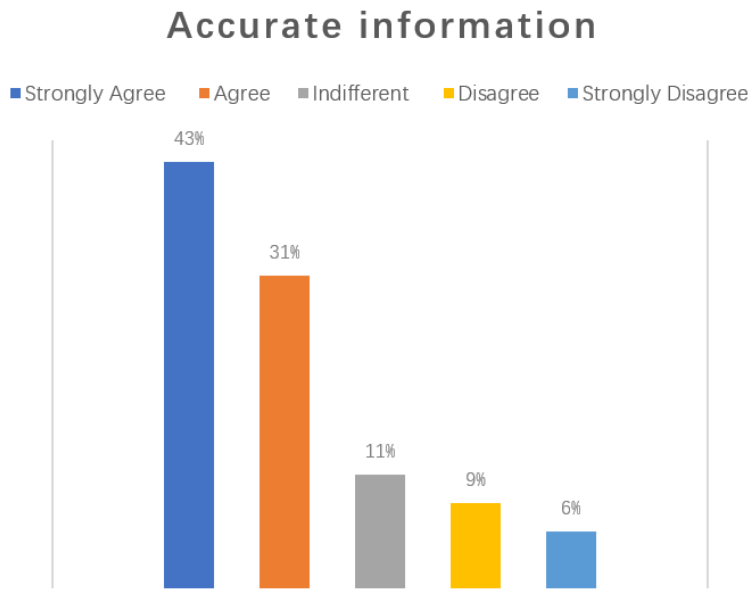


Figure 1. Accurate information

The study participants were asked whether the adoption of artificial intelligence in audit quality processes within banks results in traceable audit evidence, as shown in Figure 2. A total of 34% of the respondents strongly agreed that the adoption of artificial intelligence results in traceable audit evidence. An additional 34% of the respondents agreed to this notion. Overall, 68% of the respondents agreed that the adoption of artificial intelligence results in traceable audit evidence. A total of 14% of the respondents highlighted that they were indifferent regarding this notion. While on the other hand, a total of 18% of the respondents disagreed that the adoption of artificial intelligence results in traceable audit evidence.

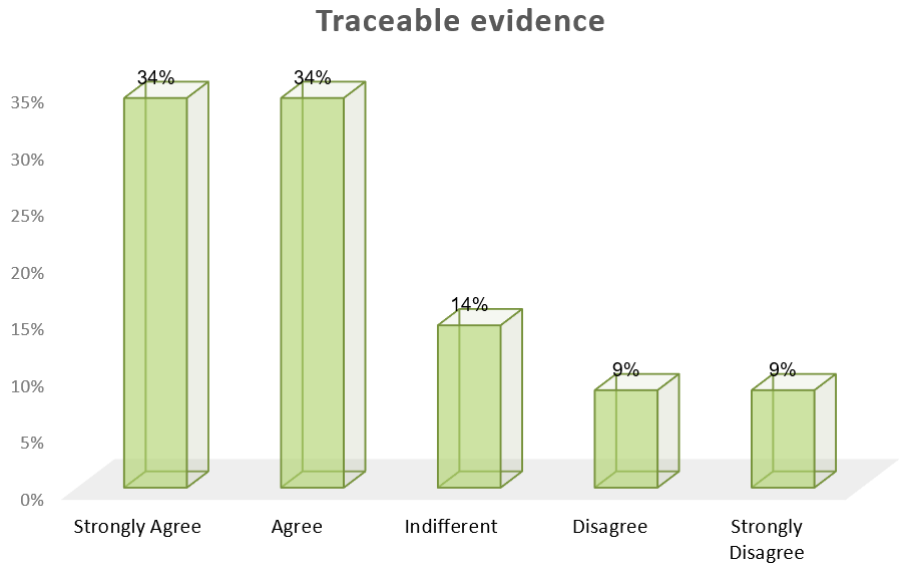


Figure 2. Traceable evidence

Figure 3 displays the responses from bank managers regarding whether the use of artificial intelligence in banks leads to more transparent audit evidence. The findings indicate that 43% of respondents strongly agreed, while an additional 37% agreed, resulting in a combined 80% expressing a positive view. Meanwhile, 11% of the participants remained neutral, and only 9% disagreed with the statement. These results suggest strong consensus among respondents that artificial intelligence contributes to greater transparency in audit evidence. This finding aligns with the work of Fırat (2025), who concluded that artificial intelligence technologies support secure and transparent transaction recording across computer networks, thus enhancing the clarity and trustworthiness of audit processes.

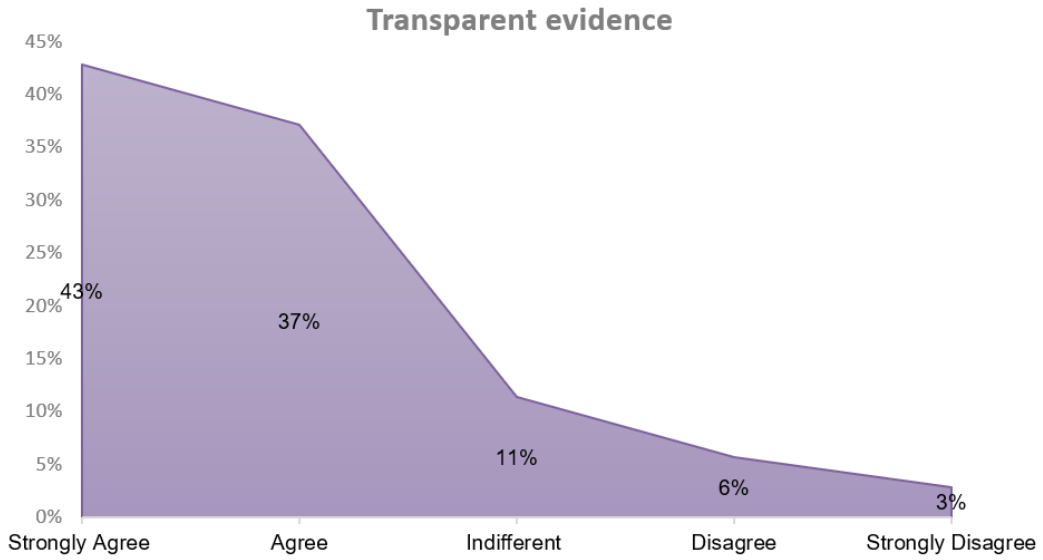


Figure 3. Transparent evidence

To assess the impact of artificial intelligence on the quality of audit evidence, participants were asked whether the adoption of artificial intelligence in banks facilitates the timely provision of audit evidence. As shown in Figure 4, the results revealed that 49% of respondents strongly agreed, while an additional 43% agreed, amounting to a total of 92% in favor of this notion. Meanwhile, 6% remained neutral, and only 2% disagreed with the statement. These findings suggest a strong consensus that artificial intelligence enhances the timeliness of audit evidence in banking institutions. This outcome supports the model's indication that artificial intelligence adoption leads to more efficient audit processes. According to Fırat (2025), artificial intelligence systems are capable of delivering real-time insights into anomalies or risks, enabling auditors to access timely and relevant audit evidence. This proactive, data-driven approach enhances audit responsiveness and allows auditors to address potential issues as they emerge, ultimately improving the overall quality and reliability of audit outcomes.

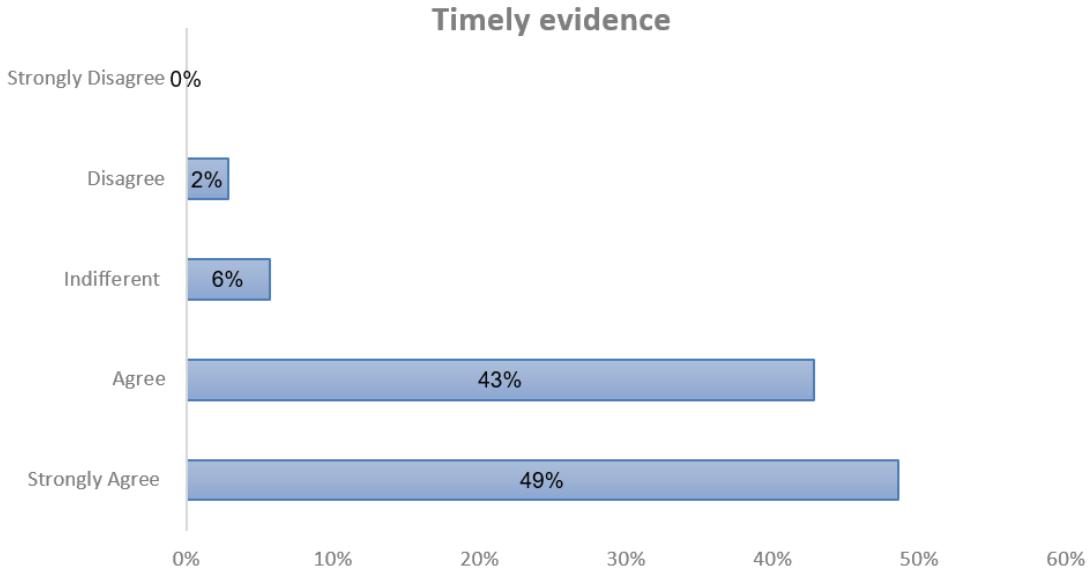


Figure 4. Timely evidence

The pie chart in Figure 5 illustrates responses from bank managers regarding whether the adoption of artificial intelligence reduces errors of omission in audit evidence. A total of 34% of respondents strongly agreed, and an additional 28% agreed with this statement, totaling 62% in agreement. A total of 20% of the managers were neutral, while 18% disagreed. Overall, the study concluded that the adoption of artificial intelligence helps reduce errors of omission in audit evidence.

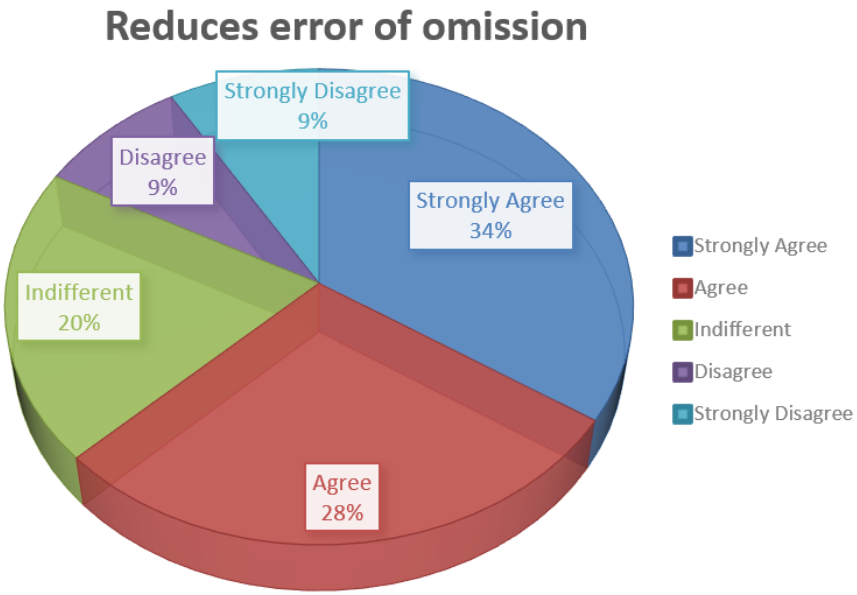


Figure 5. Minimization of omissions in financial reporting
Source: Field work.

Figure 6 presents the responses from bank managers regarding whether the use of artificial intelligence reduces calculation errors in audit evidence. A total of 40% of the respondents agreed with this statement, while an additional 37% also concurred. A total of 11% of the respondents were neutral on the matter, and 12% disagreed that artificial intelligence helps reduce calculation mistakes in audit evidence.

Figure 7 shows that 43% of respondents strongly agreed that artificial intelligence enhances the inspection of audit evidence, while an additional 40% agreed with this view. Overall, 83% of the participants supported the idea that artificial intelligence improves information inspection. Meanwhile, 9% of respondents were neutral, and 8% disagreed with the statement.

Figure 8 presents the responses from participants regarding whether adopting artificial intelligence in banks eliminates human error in audit evidence. A total of 46% of respondents strongly agreed, and another 46% agreed with this statement, totaling 92% in agreement. A total of 6% of respondents were neutral, while only 2% disagreed. The study's findings indicate that the adoption of artificial intelligence in banks effectively reduces human error in audit evidence.

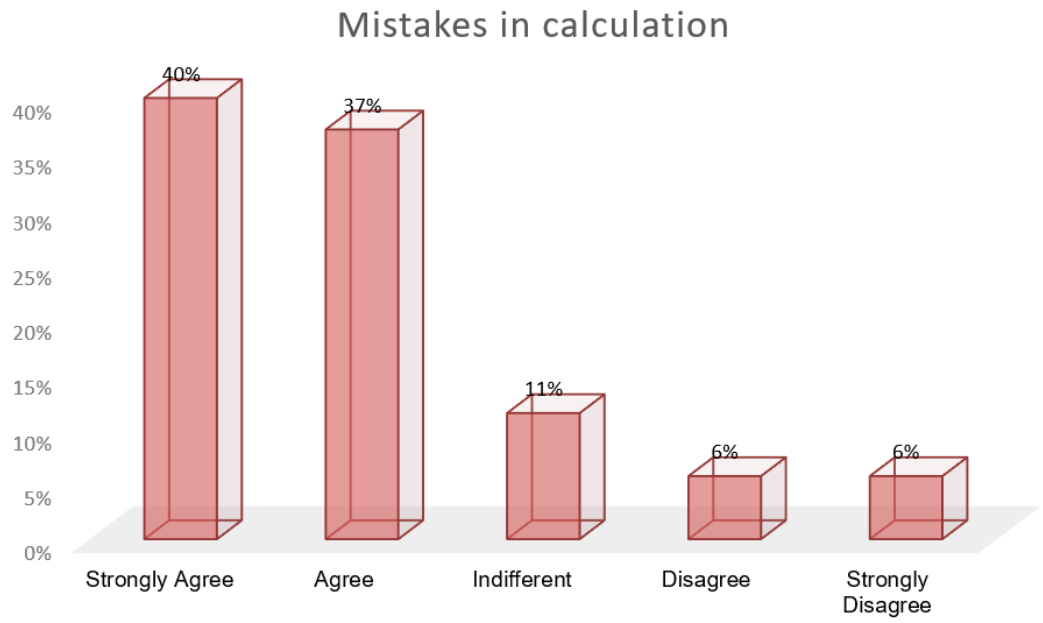


Figure 6. Minimization of calculation errors

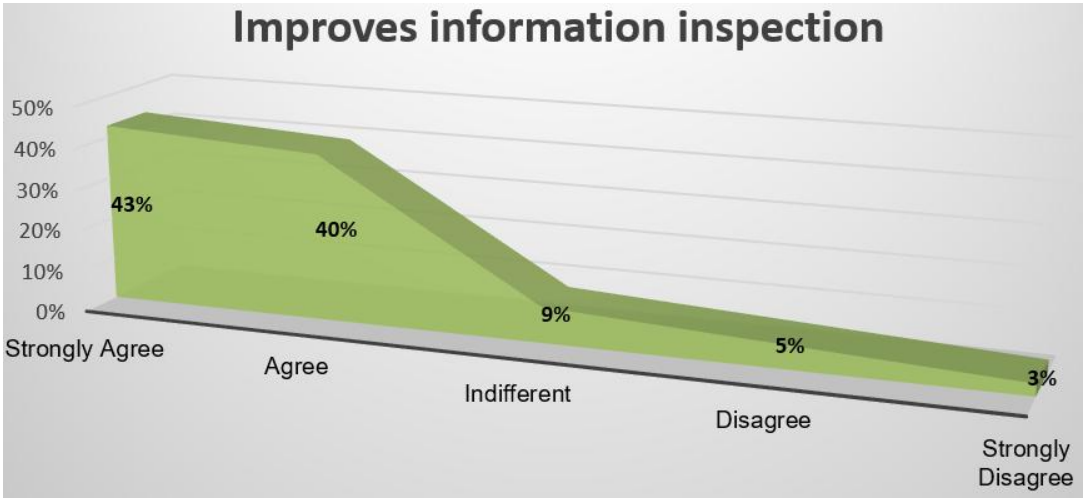


Figure 7. Enhanced examination of information
Source: Field work.

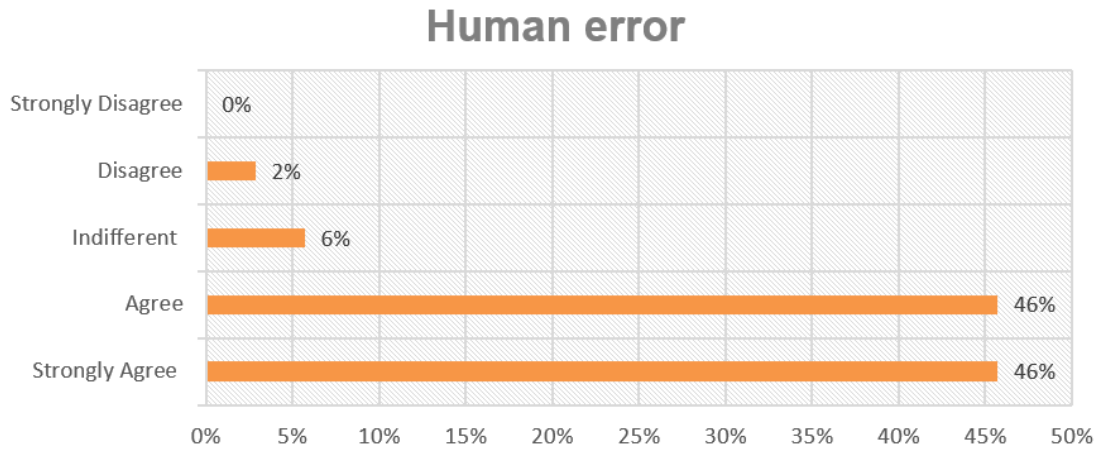


Figure 8. Minimization of human error
Source: Field work.

Figure 9 presents the responses from bank managers regarding whether artificial intelligence in banks supports comprehensive data analysis to ensure reliable audit evidence. Of the respondents, 43% strongly agreed with this statement, while an additional 37% agreed. Overall, 80% of the bank managers concurred that artificial intelligence aids in thorough data analysis to enhance audit evidence reliability. Meanwhile, 11% remained neutral, and 9% disagreed with the assertion. The findings from Figure 9 indicate that the use of artificial intelligence in banks contributes significantly to the comprehensive analysis of data, thereby ensuring more reliable audit evidence.

The participants were further asked whether adopting artificial intelligence in banks enhances the ease of cross-checking audit evidence. As shown in Figure 10, the results showed that 40% of respondents strongly agreed with this, while an additional 34% agreed, totaling 74% in agreement. Meanwhile, 14% of respondents were neutral on the matter, and 12% disagreed that artificial intelligence adoption facilitates easier cross-checking of audit evidence. Overall, the analysis indicates that implementing artificial intelligence in banks significantly improves the ability to cross-check audit evidence efficiently.

Figure 11 above presents responses from bank managers regarding whether the use of artificial intelligence in banks leads to the timely provision of audit evidence. A total of 46% of respondents strongly agreed with this statement, while an additional 43% agreed, bringing the total agreement to 88%. A total of 6% of the respondents were indifferent, and 5% disagreed with the notion. Based on these findings, the study concludes that the use of artificial intelligence in banks significantly contributes to the timely provision of audit evidence.

To determine the impact of artificial intelligence on the reliability of audit evidence, the participants were asked whether the adoption of artificial intelligence in banks leads to the consistent provision of audit evidence. As shown in Figure 12, a total of 40% of the respondents strongly agreed with this statement, while an additional 34% agreed, totaling 74% in agreement. A total of 14% of the participants were indifferent, and 12% disagreed. Based on these responses, the study concluded that the adoption of artificial intelligence in banks promotes the consistent

provision of audit evidence.

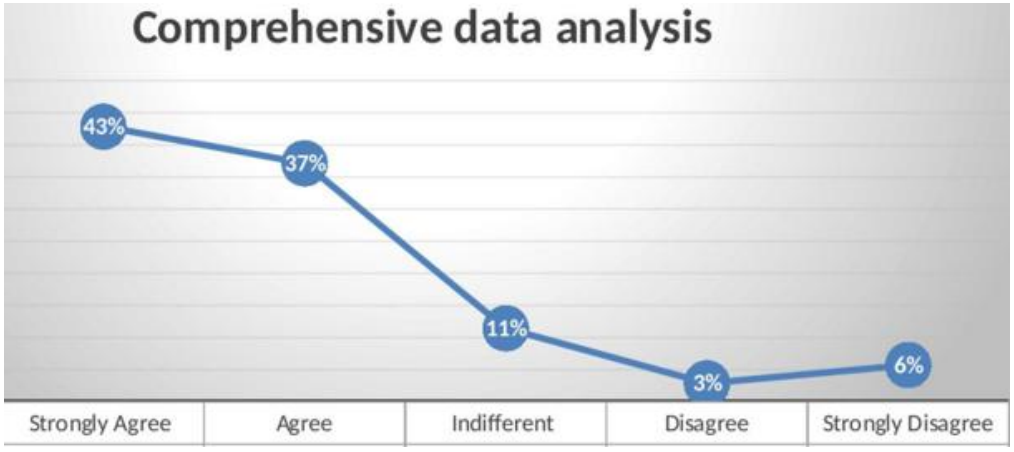


Figure 9. Comprehensive analysis of data
Source: Field work.

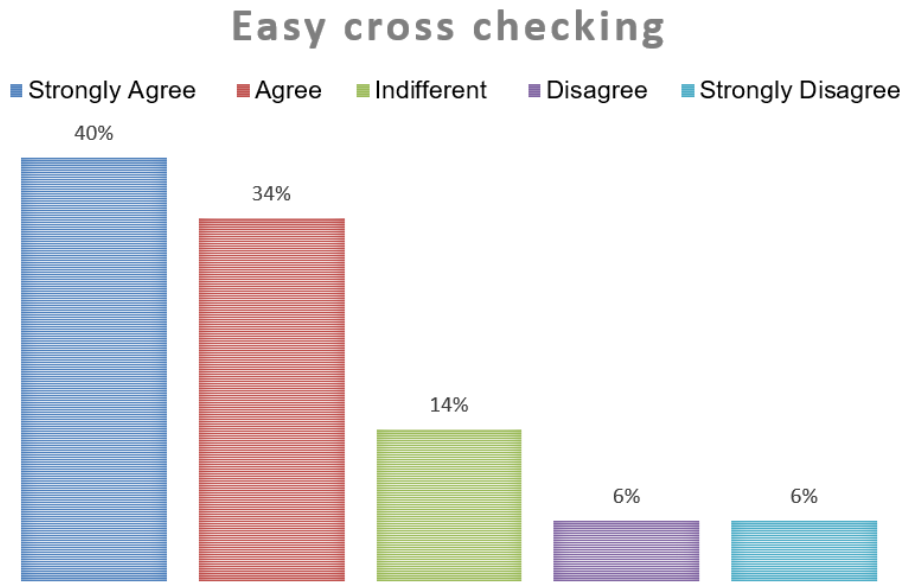


Figure 10. Easy cross-checking
Source: Field work.

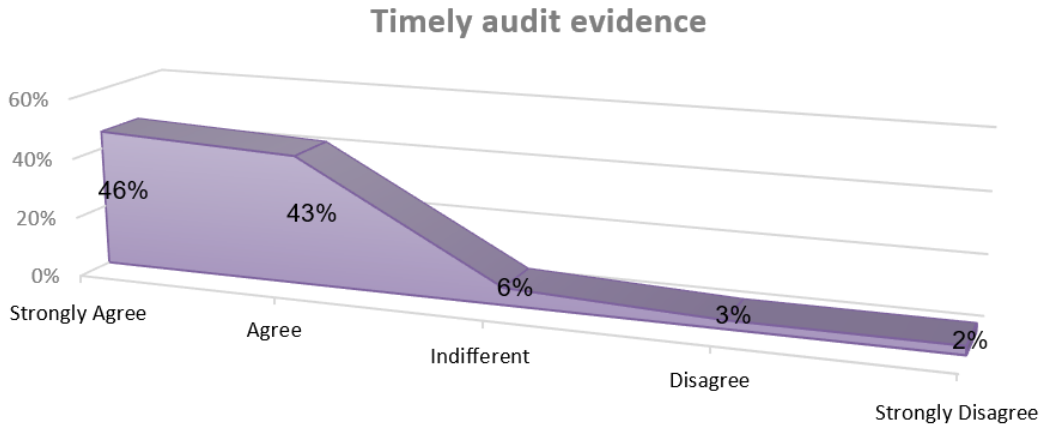


Figure 11. Timely audit evidence
Source: Field work.

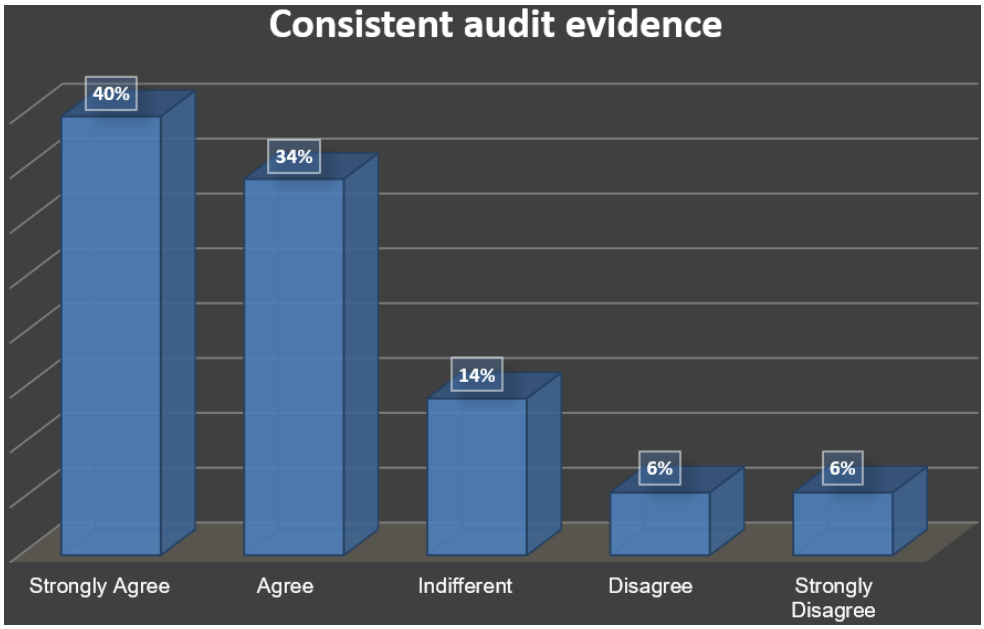


Figure 12. Consistent audit evidence
Source: Field work.

The pie chart in Figure 13 displays bank managers' responses regarding challenges in using artificial intelligence for audit quality. A total of 40% strongly agreed, and 34% agreed that the complexity of advanced artificial intelligence innovations poses a challenge to artificial intelligence use in audit evidence. Overall, 74% acknowledged this complexity as a challenge, while 14% were indifferent, and 12% disagreed. The study concluded that the complexity of advanced artificial intelligence innovations is a significant challenge in applying artificial intelligence to audit evidence.

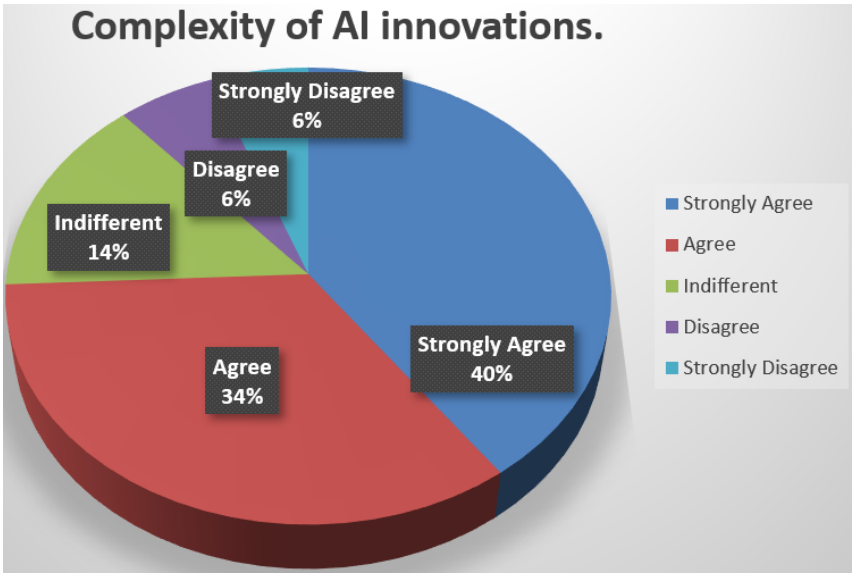


Figure 13. Complexity of advanced artificial intelligence innovations
Note: AI = artificial intelligence. Source: Field work.

Figure 14 illustrates the feedback from bank managers regarding the difficulties of implementing artificial intelligence in improving audit quality within commercial banks. A total of 80% of the participants agreed that staying updated with the latest artificial intelligence developments is a major challenge—43% strongly agreed, and 37% agreed. Additionally, 11% of the respondents were neutral on this issue, while only 9% disagreed. Based on the most common response, the study concludes that keeping pace with advancements in artificial intelligence represents a significant challenge for commercial banks in Zimbabwe.

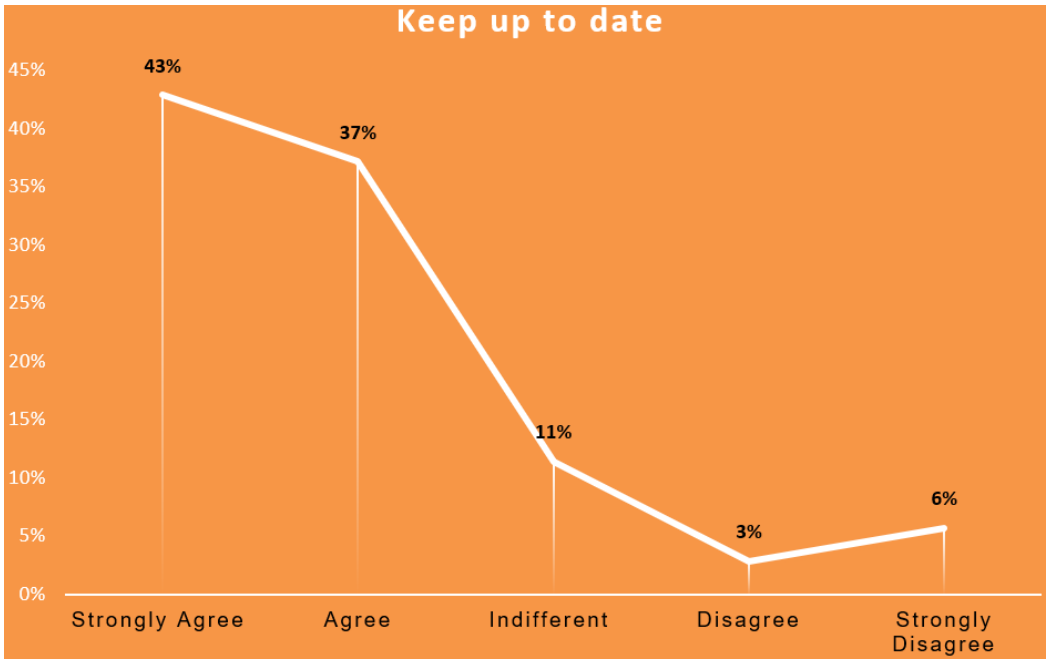


Figure 14. Challenge of staying current with artificial intelligence advancements
Source: Field work.

Figure 15 presents the responses from study participants regarding the challenges of using artificial intelligence to improve audit quality in commercial banks. A total of 89% of respondents agreed that artificial intelligence in banking is vulnerable to risks such as unauthorized access or cybercrime, with 49% strongly agreeing and 40% agreeing. Meanwhile, 9% of respondents were neutral on this issue, and only 2% disagreed. Based on the mode of responses, the study concludes that the use of artificial intelligence in banking is indeed exposed to risks related to unauthorized access and cybercrime.



Figure 15. Threat posed by unauthorized access, cyberattacks, or other forms of cybercrime
Source: Field work.

5. Discussion

The findings of this study highlight the significant impact of emerging technologies on enhancing audit quality within banking institutions. A strong majority of bank managers agree that advanced tools improve the accuracy, traceability, transparency, and timeliness of audit evidence, all critical factors for reliable financial reporting. For example, 74% of respondents indicated that such technologies help increase the precision of audit findings, while 80% acknowledged their role in facilitating comprehensive analysis of financial data. Additionally, 88% noted the importance of these tools in ensuring audit evidence is available promptly, supporting timely decision-making and compliance. These outcomes correspond with established accounting principles that emphasize the importance of accuracy and completeness in audit processes. The reduction of human errors and enhanced transparency afforded by these technologies also align with the goals of minimizing audit risk and improving the integrity of financial

statements (Fırat, 2025). Thus, technology adoption is shown to support auditors in meeting professional standards and regulatory requirements more effectively.

These findings, however, contrast with evidence from more institutionally advanced contexts, where Albawwat & Frijat (2021) found that auditors in Jordan perceived autonomous artificial intelligence systems as significantly harder to use (mean = 2.69) compared to assisted and augmented tools, suggesting that the uniformly positive perceptions reported in this study may reflect the early-stage exposure of Zimbabwean banking auditors rather than deep operational experience with artificial intelligence. Similarly, Noordin et al. (2022) reported no significant difference in artificial intelligence's perceived contribution to audit quality between local and international firms in the United Arab Emirates, a finding that challenges the assumption that institutional context drives artificial intelligence outcomes—implying that the positive results observed in this study are not unique to Zimbabwe but rather reflect a broader pattern of artificial intelligence acceptance that transcends regional boundaries, though the specific barriers identified in this study, particularly cybersecurity risks and skills shortages, remain distinctly characteristic of developing country settings.

However, the study also identifies key challenges in integrating such innovations within traditional audit frameworks. A notable concern is the complexity involved in applying new audit techniques, which may require auditors to continuously update their knowledge and skills to keep pace with technological advancements. Furthermore, risks related to unauthorized access and cybersecurity threats raise issues about safeguarding sensitive financial information during the audit process. Another important limitation highlighted is the inability of technology to fully replace human judgment and professional skepticism, which remain essential for evaluating complex accounting estimates and assessing risks. Auditors must therefore balance reliance on automated tools with critical thinking and expertise to ensure audit quality.

Overall, while the adoption of advanced audit technologies offers considerable benefits in enhancing evidence reliability and audit efficiency, addressing challenges such as cybersecurity, ongoing professional development, and maintaining human oversight is vital. These findings reinforce the need for the accounting profession to integrate technology thoughtfully while preserving core audit principles to uphold the quality and credibility of financial reporting (Fırat, 2025; McAfee & Brynjolfsson, 2017).

6. Conclusions and Recommendations

The findings clearly demonstrate that incorporating advanced technologies into banking audit processes significantly improves the quality and reliability of audit evidence. A large majority of bank managers agree that these technologies enhance key audit attributes such as accuracy, traceability, transparency, and timeliness—critical factors for trustworthy financial reporting. This aligns with existing research emphasizing the ability of such tools to efficiently analyze large volumes of financial data while minimizing human error. Their effectiveness in reducing errors of omission, calculation mistakes, and ensuring consistent, thorough data analysis underscores their potential to raise audit standards. The strong agreement among respondents indicates that these innovations are seen as transformative for audit quality, enabling auditors to produce more reliable, timely, and comprehensive evidence that supports better-informed decision-making in banking. However, the study also highlights important challenges linked to adopting these technologies in auditing. Respondents expressed concerns about the complexity of advanced audit tools, the need for continuous professional development to keep pace with technological changes, and vulnerabilities related to cybersecurity threats and unauthorized access to sensitive financial data. Furthermore, there is widespread recognition that human judgment and professional skepticism remain indispensable, as technology cannot fully substitute for auditors' expertise and nuanced decision-making in complex accounting matters. These challenges emphasize the need for strong cybersecurity protocols, ongoing auditor training, and hybrid audit approaches that combine technological capabilities with human oversight to effectively manage risks and maximize the benefits of technology in auditing.

In summary, while advanced audit technologies offer significant promise for enhancing audit quality and efficiency, addressing these challenges is essential to ensure their secure, sustainable integration into banking audits and to uphold the integrity of financial reporting. These conclusions are grounded in the descriptive statistics of this study and are further supported by survey responses. Similarly, the conclusion that artificial intelligence adoption poses significant challenges is directly evidenced, ensuring that each broad conclusion drawn in this section is traceable to a specific empirical result reported earlier in this study.

In terms of prioritized recommendations, the most pressing intervention is for the Reserve Bank of Zimbabwe to mandate minimum cybersecurity standards specifically for artificial intelligence-assisted audit systems within commercial banks, ahead of broader regulatory frameworks, as this directly addresses the most empirically prevalent barrier identified in this study. As a second priority, the banks should establish structured artificial intelligence literacy programs targeting internal auditors, with curriculum aligned to the three audit quality dimensions measured in this study—audit evidence quality, material misstatement reduction, and reliability—rather than generic technology training, thereby ensuring that capacity-building efforts are directly responsive to the specific operational gaps this study has identified.

Author Contributions

Conceptualization, O.W. and R.N.; methodology, O.W.; formal analysis, O.W. and R.N.; investigation, O.W. and R.N.; writing—original draft preparation, O.W.; writing—review and editing, R.N.; supervision, O.W. All authors have read and agreed to the published version of the manuscript.

Informed Consent Statement

Informed consent was obtained from all subjects involved in the study.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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